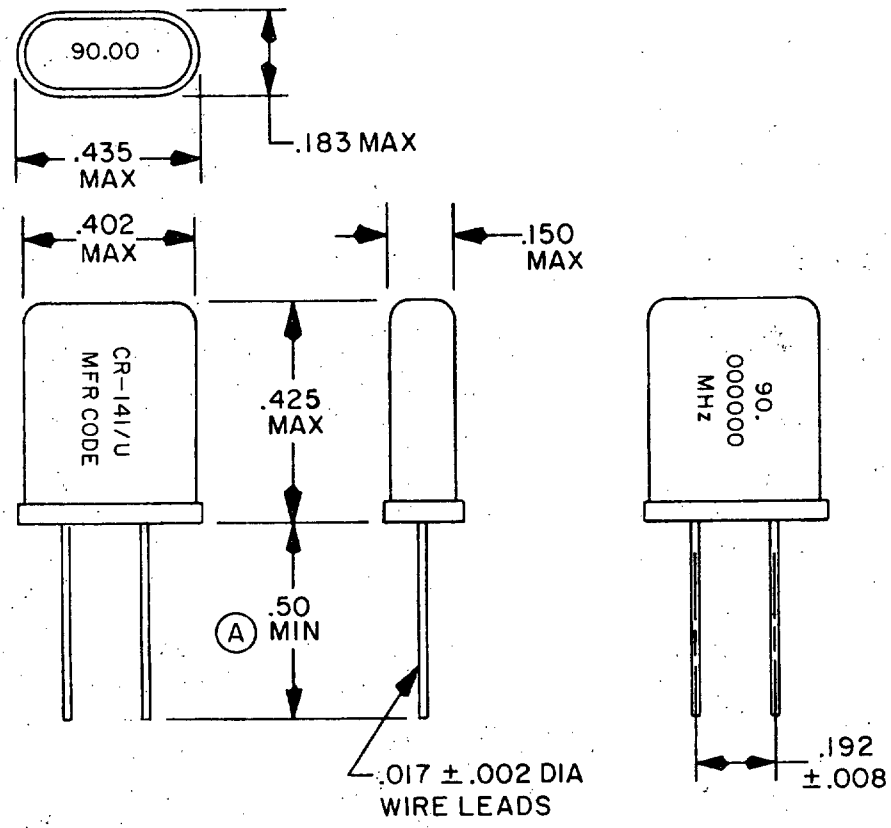
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-141/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 48 to 90 MHz; fifth overtone; noncontrolled; series resonance.



INCHES	MM
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.425	10.80
.435	11.05
.50	12.7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designator, frequency, and manufacturer's code shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Crystal unit - CR-141/U.

Ⓐ denotes changes

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FSC 5955

Ⓐ TABLE I. Requirements.

Requirement	Specified value or condition
Holder	Modified HC-18/U or HC-43/U (MIL-H-10056/9) See figure 1.
Dimensions, marking, and configuration	See figure 1.
Frequency range	48 to 90 MHz, inclusive
Capacitance, shunt	7.0 pF, maximum
Mode of oscillation	Fifth Mechanical Overtone
Government reference standard test set	TS-683/TSM
Rated drive level	2.0 ±0.4 mW
Calibration values:	
Resistance	50 ohms
Resistance voltage drop	0.32 volt
Operating temperature ranges:	
Primary	-30°C to ±55°C, inclusive
Secondary	-55°C to -30°C and 55°C to 65°C, inclusive
Frequency tolerances:	
Primary operating temperature range	±25 ppm
Secondary operating temperature range	±45 ppm
Equivalent resistance	50 ohms, maximum
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±5 ppm

1/ Method 204 of MIL-STD-202, test condition A

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11
DLA - ES

User activities:
Air Force - 17

Agent:
DLA - ES

Preparing activity:
Army - ER

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